

电梯门机控制器电磁兼容测试探讨 *

唐海龙,王会方,李晓露
南京市特种设备安全监督检验研究院

摘 要 | 为保障电梯稳定安全运行,对电梯门机控制器进行综合的电磁兼容测试十分必要。首先介绍电梯门系统的构成及电梯门机控制器的工作原理,说明电梯门机控制器在电梯门系统中的重要性。其次,针对电梯门机控制器进行电磁兼容试验提出一套工况模拟的测试平台,并依据两个针对电梯本体及部件的现行电磁兼容抗扰度及发射标准,对一款门机控制器电磁兼容试验项目内容进行确定。最后,对电梯门机控制器电磁兼容试验中的注意事项及干扰进行分析。

关键词: 电梯; 门机控制器; 电磁兼容

随着城市化进程的推进,电梯保有量持续增加,随之而来的电梯故障总量也开始增多。据统计,在所有的电梯故障中,电梯门系统故障约占 70%^[1]。电梯的运行过程中门系统动作频繁而且与乘客接触密切,电梯门机控制器作为门系统的核心部件,其工作运行的安全可靠、功能的稳定性尤为重要。在各类影响因素中,电磁兼容影响是一类十分关键的影响因素^[2],因此,在完善电梯门机控制器运行的可靠性方面,针对电梯门机控制器进行电磁兼容测试十分必要。

1 电梯门系统的构成

电梯门系统主要由门机控制器、门机电动机(含编码器)、门扇、驱动机构及附件、电气安全装置等构成^[3]。电梯门机电动机分为直流电机、交流异步电机、永磁同步电机。其中,永磁同步电机具有体积较小、动态性能高、调速

范围大的优点,在电梯门机系统中应用广泛。电梯门扇有旁开式、中分式以及垂直滑动式,其中以中分式门扇使用最多。驱动机构有机械连杆、钢丝绳、皮带等形式的连接方式。电梯门扇的开门、关门动作由门机控制器驱动门机电机正向或反向转动,产生动力驱动机械连接部件实现电梯门扇在门导轨上往复运动。图 1 是某型号电梯门系统实物图,图 2 为某公司生产的一款电梯门机控制器。

图 3 是电梯门机控制系统框图。电梯门机控制器是电梯门系统的“大脑”,它接收电梯主控制器的开关门指令,收集电梯门系统门区信号、关门到达信号、光幕信号等,综合采集到信息后控制驱动信号使门机电机工作以及输出故障信号,实现处理各种情况下的安全开关电梯门的功能,并将门的状态数据返回给电梯主控制器。有的电梯门机控制器还具有自学习功能,能够适配不同门扇

* 基金项目: 江苏省市场监督管理局科技计划项目 融合数字孪生与混合现实的电梯远程诊断及救援指引技术研究 (KJ2023039)

作者简介:唐海龙(1989—),男,安徽宣城人,工程师,硕士,研究方向为机电类特种设备产品检验检测技术。

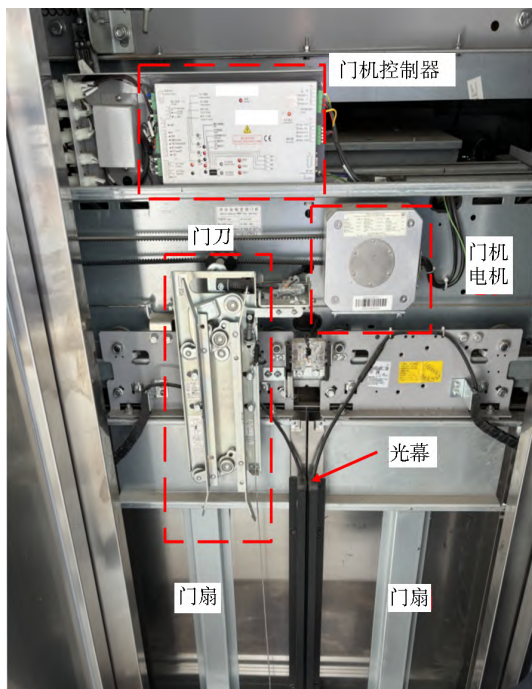


图1 电梯门系统



图2 电梯门机控制器

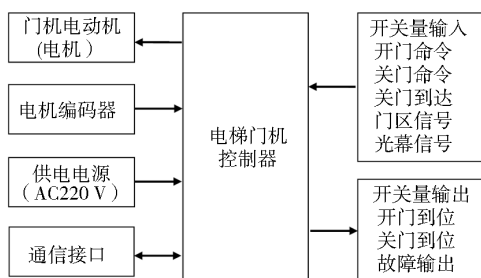


图3 电梯门机控制系统框图

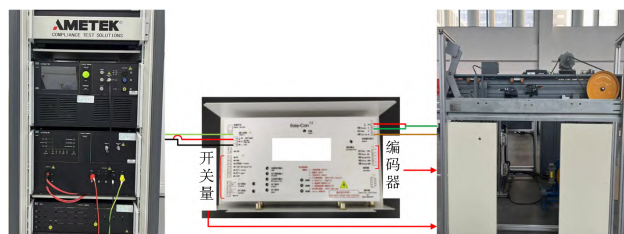
宽度的电梯门系统。电梯门机控制器大都采用编码器获取门机电机速度和位置,再辅助开门到位、关门到位、

门扇宽度等信号参数获取门扇的绝对位置,可以使电机运行更为精准,提高电梯的运行质量^[4]。

2 电梯门机控制器电磁兼容测试系统

2.1 电梯门机控制器测试平台

鉴于电梯门系统在整个电梯装备里工作的重要性,以及电梯门机控制器在电梯门系统的核心作用,针对电梯门机控制的电磁兼容的试验测试必不可少。为实现电梯门机控制器电磁兼容性的测试,需要搭建一个实现模拟电梯门机控制器正常工作各类状态的试验平台,在某些测试项目中能够使电梯门机控制器工作在最不利的工作状态下。在此状态下进行电磁兼容项目的测试,可以最大限度地反应出电梯门机控制器性能的极限,从而使电磁兼容项目的测试结果更为准确。图4为电梯门机控制器电源线高能浪涌的测试平台布置示意图。测试时,使用工况模拟装置使电梯门机控制器运行在最不利工作状态下,测试设备通过其内部的电源耦合网络将高能浪涌干扰叠加在输出电源上,在给电梯门机控制器供电的同时施加干扰。



(a) 三相综测仪 (b) 门机控制器 (c) 门机工况模拟装置
图4 电梯门机控制器电磁兼容测试平台

2.2 屏蔽室、暗室的电磁兼容测试

在非屏蔽室、非暗室的测试场景的电磁兼容项目中,只需要搭建起前述电梯门机控制器类似的测试平台,使电梯门机控制器工作在模拟工作状态下,对电梯门机控制器施加对应电磁兼容项目的干扰,观察测试平台有无异常情况发生。针对在屏蔽室、暗室进行的测试项目,则需要对测试平台做出改动:只将被测样品——电梯门机控制器放置在屏蔽室或暗室中,将测试平台中可能影响测试结果的其他部分放置到屏蔽室或暗室外,具体布置方式见图5。内外电路的连接应通过专用的连接接口,必要情况下还需使用滤波器器件。

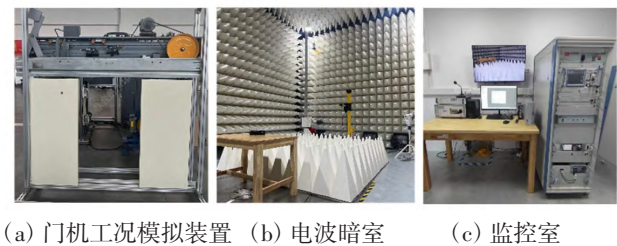


图5 电梯门机控制器电磁兼容布置

3 电梯门机控制器电磁兼容测试内容

3.1 抗扰度测试

针对电梯门机控制器进行电磁兼容测试需要考虑的情况之一是在受到外部电磁干扰的情况下，测试电梯门机控制器工作运行的稳定性。这就需要对电梯门机控制器进行电磁兼容抗扰度（EMS）测试。在标准 GB/T 24808—2022《电磁兼容 电梯、自动扶梯和自动人行道的产品系列标准 抗扰度》^[9]（以下简称 EMS 标准）中，对于电梯门机控制器进行电磁兼容抗扰度（EMS）测试的测试项目有静电放电（ESD）、电快速瞬变脉冲群（EFT）、浪涌（Surge）、电压跌落（DIP）、辐射（RS）、传导抗干扰（CS）。

针对图 2 所示电梯门机控制器，分析其结构及基本的运行工作原理，得出该款电梯门机控制器需要进行 EMS 试验的端口有外壳端口、供电电源端口、电机端口、开关量输入输出端口以及编码器端口。再依据 EMS 标准中的要求，针对不同端口选取对应的试验项目及等级进行 EMS 试验。表 1 列出了该款电梯门机控制器不同端口的 EMS 项目，“√”表示实施该项目，“—”表示不实施该项目。其中，外壳端口会接触到操作人员以及起到隔离电磁场影响的作用，需要对其进行静电放电、射频电磁场试验。供电电源端口为交流供电，适用于 EMS 标准中表 6 的试验项目，有电快速瞬变脉冲群共模（EFT）、电压跌落（DIP，包括电压暂降和电压中断）、浪涌（Surge）、射频场感应的传导骚扰共模（CS）4 个测试项目。门机电机端口为三相交流端口，适用于 EMS 标准中表 6 的试验项目，有电快速瞬变脉冲群共模（EFT）、浪涌（Surge）、射频场感应的传导骚扰共模（CS）3 个测试项目。开关量输入输出端口以及编码器端口，应该参照 EMS 标准中表 2 来确定试验项目，有电快速瞬变脉冲群共模（EFT）、射频场感应的传导骚扰共模（CS）2 个测试项目。

表 1 电梯门机控制器 EMS 测试端口项目表

端口	ESD	EFT	Surge	DIP	CS	RS
外壳	√	—	—	—	—	√
电源	—	√	√	√	√	—
电机	—	√	√	—	√	—
开关量	—	√	—	—	√	—
编码器	—	√	—	—	√	—

3.2 发射测试

电梯是一个整体的机电产品，电梯门机控制器作为电梯产品里面的一个部件，其工作时产生的电磁干扰水平也需要纳入其整体电磁兼容水平的考虑范围。一旦其产生的电磁干扰水平（发射水平：EMI 水平）超出标准限定的范围，势必会影响到其周围的其他电梯电子部件的正常运行功能，从而导致电梯故障发生的概率增加。在标准 GB/T 24807—2021《电磁兼容 电梯、自动扶梯和自动人行道的产品系列标准 发射》^[9]（以下简称 EMI 标准）中，对于电梯门机控制器进行电磁兼容发射（EMI）测试的测试项目有辐射发射（RE）、传导发射（CE）、谐波发射（Harmonic）。

针对图 2 所示电梯门机控制器，需要对其进行 EMI 试验的端口有外壳端口、供电电源端口、电机端口。其中电机端口由于电缆长度小于 2 m，无需进行相关测试。依据 EMI 标准的要求，选取相应的发射限值对试验测得的限值进行评判。外壳端口的辐射发射测试是如果当电梯门机控制器内部的具有超过辐射发射限值的电子器件时，外壳能否将整体的发射水平限制在限值范围内。这主要考量的是外壳端口开头的形式、位置和大小是否能满足要求。电源线端口主要测试电梯门机控制器通过电源线向电梯中的其他部件传导电磁干扰以及谐波电流。表 2 为该款电梯门机控制器 EMI 测试端口项目表，“√”表示实施该项目，“—”表示不实施该项目。

4 电梯门机控制器电磁兼容测试分析

4.1 电磁兼容测试注意事项

（1）样品的布置要符合对应标准以及相应电磁兼容项目作业指导书的要求，熟悉被测样品的基本工作原理，

表2 电梯门机控制器 EMI 测试端口项目表

端口	RE	CE	Harmonic
外壳	√	—	—
电源口	—	√	√

熟悉仪器设备的操作流程,在浪涌试验等具有高压危险情况时注意个人安全防护。

(2) 根据标准要求以及用户需要选择与被测样品相适应的测试等级以及性能判据。有时用户会根据样品应用的特殊使用环境,要求以高于产品标准的试验等级来测试,需要注意测试等级的选用以及性能判据的选择。

(3) 做好试验记录尤其是异常情况出现时的记录,确保可以现象复现,为找出异常出现的原因,提供充足的分析及原因排除资料,也为用户整改问题提供充足的帮助。

4.2 电磁兼容干扰分析

产品的电磁兼容性能是产品整体设计能力的体现,就电梯门机控制器而言,主要有以下3个方面容易产生电磁兼容干扰问题:

(1) 外壳端口。外壳作为设备的物理边界,其设计和材料选择对于电磁屏蔽效果至关重要。外壳端口能够提供辐射和静电放电能量的传播途径,合理的外壳设计可以有效地阻挡外部电磁波的进入,减少电磁干扰对设备内部电路的影响。同时,外壳端口也能防止设备内部产生的电磁辐射泄漏到外部环境中,避免对其他设备造成干扰。

(2) 电源端口。电梯门机控制器的电源主要起两部分的作用:一是要通过 AC-DC 转成直流电给整个控制器供电;另外是需要给电梯门机电机供电。容易产生传导发射、辐射发射超标等问题。主要是由电源电路中的开关功率管、高频变压器、整流二极管等器件引起,需要电源设计者通过合理的器件选择、合理的前端滤波器设计、有效的屏蔽措施、良好的接地处理来抑制传导发射和辐射发射的超标。

(3) 开关量端口。开关量作为可直接进入芯片内部的端口在电磁兼容试验中容易产生电快速瞬变脉冲群、浪涌等抗扰度不足的问题,这主要是信号未充分隔离的原

因。一般采用输入端口用光耦隔离、输出端用继电器隔离的方式将施加的干扰隔离在外侧的电路中,里侧再用滤波电路将通过“爬电现象”过来的干扰滤掉。同时,需要保证输入端子公共端的接地回路良好接地。

5 结束语

随着国内电磁兼容相关水平的提升,电梯行业针对电梯相关电子部件的电磁兼容水平的状况也越来越重视,电梯电子部件的电磁兼容测试需求会愈加旺盛,本文主要阐述了电梯门机控制器在电梯门系统中的关键性,针对门机控制器电磁兼容测试的平台布置、测试内容、注意事项及干扰分析,可为电梯门机控制器及其他电梯电子部件进行电磁兼容测试提供一定参考。

参考文献

- [1] 郭淼,潘锋,李斌,等.基于大数据的电梯检验检测技术认可及故障分析应用研究[J].中国特种设备安全,2023,39(7):63-67.
- [2] 王成华.电梯电气系统的电磁兼容性[J].电子技术与软件工程,2017(15):87.
- [3] 王俊龙.浅谈电梯门系统智能检测的应用[J].中国电梯,2018,29(15):71-72.
- [4] 沈夏,章睿.浅谈家用电梯门机控制系统的设计[J].中国电梯,2023,34(2):29-30.
- [5] GB/T 24808—2022 电磁兼容 电梯、自动扶梯和自动人行道的产品系列标准 抗扰度[S].
- [6] GB/T 24807—2021 电磁兼容 电梯、自动扶梯和自动人行道的产品系列标准 发射[S].

通信地址:江苏省南京市浦口区万寿路15号 南工大科技产业园(210001)

(收稿日期:2024-08-27)

control machine for mass concrete can automatically and intelligently control the temperature, flow rate and flow direction of the cooling circulating water according to the pre-set parameters, which greatly improves the efficiency of temperature control and anti-cracking of mass concrete and achieves the purpose of minimizing concrete cracking.

Keywords: Mass concrete; Cooling circulating water; Temperature stress; Intelligent control; Crack control

Development and Research of Intelligent Sports Chassis Based on ADAS Testing

Intelligent sports chassis, as the main testing equipment for intelligent driving testing, needs to carry various soft targets to perform motions of various road user scenarios, such as straight travelling, turning and lane changing. In view of the defects such as low reliability, high radar attributes and unstable motion performance of the intelligent sports chassis during the testing of active safety and automatic driving of automobiles, a structural form of adopting a multi-motor scheme to realize rear-wheel drive + front-wheel steering + one-piece multi-grid is designed, and low radar attributes are achieved by the design of beveled + circular arc surfaces of key parts of the body as well as the coating of wave-absorbing coatings; at the same time, supplemented with the multi-layer control architecture of the upper computer control system and the central domain control system, the high-precision and high-speed motion control as well as safety and stability of the intelligent sports chassis are accom-

plished through the control algorithm. The results show that, the developed intelligent sports chassis is a sports chassis with high reliability, high speed and high precision, which can travel at a high speed according to the automatically planned path.

Keywords: Intelligent sports chassis; Steering system; Drive system; Control algorithm

Analysis and Optimization of Heat Dissipation Performance of Loader Engine Compartment Based on Virtual Simulation and Test

By building a wind tunnel bench test system and a virtual wind tunnel numerical calculation model, the heat dissipation flow field of a loader engine compartment is studied, and the heat dissipation volume is optimized by adjusting the sinking amount of the fan in the windshield. The research results show that, the heat dissipation airflow increases linearly with the increase of fan speed; as the fan sinks from -30 mm to 120 mm, the total airflow passing through the radiator continues to decrease, and the heat exchange area and installation position of the radiator shall be considered comprehensively in the design of the heat dissipation system; compared to the bench test, the virtual wind tunnel numerical calculation method can be used to analyze and optimize the interior of the engine compartment in a more comprehensive and efficient way.

Keywords: Loader; Heat dissipation performance; Wind tunnel; CFD; Structural optimization

Research on Shock-Absorbing Performance of Aircraft Landing Gear Based on Joint Simulation

Taking the main landing gear of an unmanned aircraft as the research object, a joint simulation model based on multi-body dynamics software and hydraulic simulation software is established to improve the design efficiency. Considering the flexible deformation of the main landing gear strut, the landing gear strut is flexibilized. The drop test results are compared with the joint simulation results. The comparison results show that, when simulating the drop process of the toe-out main landing gear structure, the simulation results considering the flexible deformation of the strut are in good agreement with the drop test results. The influence of the initial inflation volume of the shock absorber and the positive stroke snubbing hole on the vertical load is analyzed, and the results show that both the initial inflation volume and the positive stroke snubbing hole have a significant impact on the vertical load.

Keywords: Rigid body dynamics; Hydraulic simulation; Landing gear; Shock absorber; Flexibility; Drop test

Discussion on Electromagnetic Compatibility Test of Lift Door Operator Controller

To guarantee the stable and safe operation of the lift, it is necessary to carry out a comprehensive electromagnetic compatibility test on the lift door operator controller. Firstly, the composition of the lift door system and the working principle of the lift door operator controller are introduced, and the importance of the lift door operator controller in the lift

door system is explained. Secondly, a set of test platform for working condition simulation is proposed for the electromagnetic compatibility test of the lift door operator controller, and the content of electromagnetic compatibility test items for the door operator controller is determined based on two existing standards on electromagnetic compatibility immunity and emission for lift body and components. Finally, the precautions and disturbances in the electromagnetic compatibility test of the lift door operator controller are analyzed.

Keywords: *Lift; Door operator controller; Electromagnetic compatibility*

Construction of Three-Dimensional Model of Asphalt Mixtures Based on TWS

To improve the accuracy of the three-dimensional model of asphalt mixtures, a method of combining computed tomography (CT) technology with Trainable Weka Segmentation (TWS) is used to construct a three-dimensional inhomogeneous model of asphalt mixtures. The continuous cross-section image inside the asphalt mixture is obtained through industrial CT scanning equipment, the image segmentation is processed by TWS method, and the obtained image information is imported into MATLAB to reconstruct the three-dimensional inhomogeneous model of the asphalt mixture. The accuracy of the model is verified by comparing the volume proportions of different components in the asphalt mixture specimen and in the internal structure of the three-dimensional model. The research results show that the three-dimensional model reconstructed

by this method can truly reflect the meso-structure of asphalt mixtures.

Keywords: *CT technology; Asphalt mixture; Three-dimensional reconstruction*

Robot Workspace and Finite Element Analysis Based on Spiral Theory

A flexible installation robot that can perform tasks such as lifting and installing large structural parts in the construction field is developed to undertake the assembly of structural parts. To accurately evaluate the capability of the robot, based on spiral theory, kinematic modelling of the robotic arm is carried out through exponential product equation to obtain the kinematic equation of the robotic arm, and the workspace of the robotic arm is solved. The robotic arm model is meshed, and the Workbench module is used to solve the robotic arm model to evaluate the stiffness performance of the robot, which provides a basis for the structural optimization of the robotic arm and the optimization of the load capacity of the robot under the operational conditions such as gripping and installation.

Keywords: *Load capacity; Workspace; Spiral theory; Finite element analysis*

Layout Analysis of Excavator Display Based on Differential Evolution

The visibility criterion of man-machine layout is applied to the optimization of cab layout, and the display layout is studied from a visual perspective. With the purpose of improving the efficiency of visual search and determining the position and size of the display, the visual criterion of the display

is transformed into a constraint to limit the position of the display, a mathematical model is constructed, and the differential evolution algorithm is applied to solve the model to obtain the positional relationship between the display and the eye point and to determine the spatial position of the display while giving the recommended size of the display, thereby obtaining the layout design plan of the display. The results of man-machine simulation analysis of the cab show that all the improved devices in the cab are within the reachable range and meet the requirements of man-machine visibility, which verifies the correctness of the mathematical model and algorithm solution.

Keywords: *Man-machine engineering; Display; Differential evolution; Simulation analysis; Layout design*

Structural Strength Analysis of Drive Wheels of Heavy-Duty Tracked All-Terrain Rescue Equipment

The drive wheel system is a key component of heavy-duty tracked all-terrain emergency rescue equipment, which not only bears large track tension, but also bears huge impact force from the ground during vehicle travelling, so ensuring its safe operation is the key to ensuring the normal operation of the heavy-duty tracked all-terrain emergency rescue equipment and guaranteeing the safety of passengers. Based on typical working conditions of the heavy-duty tracked all-terrain rescue equipment, the force situation of its drive wheel system is analyzed, the stress load on the bolted parts on both sides of the drive wheel system is deduced according to